

Setup Definitions

Data

- Data $\mathbf{x}$ is distributed like i.i.d Rademacher in all entries except the third entry, and the target function is $y(\mathbf{x}) = -x_1x_2$, the XOR on the first two entries
- The third entry is like a planted answer, with probability $1 - \lambda$ it is the XOR of the first two entries, and with probability λ it's the negation
- We decompose $\mathbf{w} = \mathbf{w}_{\text{sig}} + \mathbf{w}_{\text{opp}} + \mathbf{w}_{\text{sp}} + \mathbf{w}_{\perp}$ where sig is the projection onto the signal direction, which depends on the sign of a , $\mathbf{w}_{\text{opp}}$ points in the “other” direction (discussed below), $\mathbf{w}_{\text{sp}}$ is the projection onto the third coordinate, and $\mathbf{w}_{\text{perp}}$ is everything else
 - The sign of a determines the sign of a particular neuron's contribution, and therefore whether we want to align with $\pm\mu_1$ (the choices of $\mathbf{z}$ that are 1 under XOR) vs $\pm\mu_2$ (the -1 inputs) so $\mathbf{w}_{\text{opp}}$ is just the orthogonal direction to the signal
- We decompose $\mathbf{x} = \mathbf{z} + \mathbf{s} + \boldsymbol{\zeta}$, which are the projections onto the first two coordinates (so $y(\mathbf{x}) = y(\mathbf{z})$), the projection onto the third coordinate, and the remaining $d - 3$ coordinates respectively

Loss

- The neural network output $f_\rho(\mathbf{x})$ is the average of the neuron outputs, so

$$f_\rho(\mathbf{x}) = \frac{1}{p} \sum_{a_j, \mathbf{w}_j} a_j \sigma(\mathbf{w}_j^\top \mathbf{x}) = E_{(a_j, \mathbf{w}_j) \sim \rho} [a_j \sigma(\mathbf{w}_j^\top \mathbf{x})]$$

- The loss, as a function of $\mathbf{x}$ denoted $\ell_\rho(\mathbf{x})$, is twice the negative log of the sigmoid of the margin, where we let $\gamma(\mathbf{x}) := f_\rho(\mathbf{x})y(\mathbf{x})$ denote the margin, $\psi(\gamma) = \frac{1}{1 + \exp(-\gamma)}$ denote the sigmoid, and so $\ell_\rho(\mathbf{x}) = -2 \log(\psi(\gamma(\mathbf{x}))) = h(\gamma(\mathbf{x}))$ where we use h to denote the outer functions, so $h(\gamma) := -2 \log(\psi(\gamma))$ is the sigmoid composed with twice the negative log.
- We let $\ell_0(\mathbf{x})$ denote the first order approximation of ℓ_ρ around 0 with respect to $f_\rho(\mathbf{x})$, observe that because h is scaled by 2 we have $h'(\gamma) = \frac{-2e^{-\gamma}}{1+e^{-\gamma}}$ so $h'(0) = -1$, giving

$$\ell_0(f_\rho(\mathbf{x})) = \ell_\rho(0) + h'(0)\gamma(\mathbf{x}) = -2 \log \frac{1}{2} - y(\mathbf{x})f_\rho(\mathbf{x})$$

- L_ρ and L_0 are the corresponding population losses, so the expectation over $\mathbf{x}$ drawn from the described distribution
- We write $\ell_\rho^{(1)}(\mathbf{x}) = h'(\gamma(\mathbf{x}))$ for the first derivative of the loss with respect to the margin, and $\ell_0^{(1)}(\mathbf{x})$ as the corresponding value for ℓ_0 which is -1 . So I believe we're writing it as a function of $\mathbf{x}$ but really are considering the change in the loss wrt the margin, later this will be useful in bounding $\nabla(L_\rho - L_0)$.
- $\nabla_{\mathbf{w}} L$ and $\partial_a L$ are scaled by p so that we get invariance over the width of the neural network (the p cancels with the $\frac{1}{p}$ in the averaging over neurons)

Section B: L_0 Analysis

We want to characterize the gradients under the L_0 approximation of the loss to derive recurrences on each part of $\mathbf{w}$. We will ultimately show that the L_0 approximation is fine for small networks (small absolute value/norm of $(a, \mathbf{w})$), and that both $\mathbf{w}_{\text{sig}}, \mathbf{w}_{\text{sp}}$ grow according to a geometric recurrence, but that $\mathbf{w}_{\text{sp}}$ has a headstart like $\sqrt{d}$.

Lemma 0.1 (L_0 gradient along signal and opposite directions): For any neuron $a, \mathbf{w}$ we have

$$\begin{aligned}\langle -\mathbf{w}_{\text{sig}}, \nabla_{\mathbf{w}} L_0 \rangle &= \frac{\sqrt{2}}{4} |a| \|\mathbf{w}_{\text{sig}}\| \Pr_{\zeta} \left[|\langle \mathbf{w}, \zeta \rangle| + |\langle \mathbf{w}, e_3 \rangle| \leq \sqrt{\|\mathbf{w}_{\text{sig}}\|} \right] \\ \langle -\mathbf{w}_{\text{opp}}, \nabla_{\mathbf{w}} L_0 \rangle &= \frac{-\sqrt{2}}{4} |a| \|\mathbf{w}_{\text{sig}}\| \Pr_{\zeta} \left[|\langle \mathbf{w}, \zeta \rangle| + |\langle \mathbf{w}, e_3 \rangle| \leq \sqrt{\|\mathbf{w}_{\text{sig}}\|} \right]\end{aligned}$$

Okay so roughly speaking we characterize the gradient of the L_0 approximation of the population loss, we can think of the right hand side as being our gradient step up to scaling so the $\mathbf{w}_{\text{sig}}, \mathbf{w}_{\text{opp}}$ directions of $\mathbf{w}$ grow in opposite directions with the same magnitude.

Section C: L_ρ Analysis

- First shows a bound on the difference in the gradients between L_0 and L_ρ , coordinate by coordinate
- Then we reuse results from section B to characterize the gradients in $\{\mathbf{w}_{\text{sig}}, \mathbf{w}_{\text{opp}}, \mathbf{w}_{\text{sp}}, \mathbf{w}_\perp\}$ under L_ρ , but with extra steps to handle the margin term that results from us considering $\ell_\rho^{(1)}$ rather than $\ell_0^{(1)} = -1$. All lemmas rely on a certain assumption on the magnitude of the margin, and condition on a high probability event $\mathcal{E}_{\text{test}}$, generally bounding via a Lipschitzness argument.

Section D: Phase I Induction

Preliminaries

- The first lemma shows that if we get alignment between the signs of $a, \mathbf{w}_{\text{sp}}$ then and $\mathbf{w}_{\text{sp}}$ is sufficiently large compared to the other components then the network predicts based on $\mathbf{w}_{\text{sp}}$
 - a determines sign, so if it matches $\mathbf{w}_{\text{sp}}$ then $\mathbf{w}_{\text{sp}}$ determines sign, so we get no contribution when its sign differs from x_3 and otherwise we get a contribution matching x_3

Scalings

1. $\log \log(d) d^{-C} \ll \eta \ll \log^{-3}(d)$
 - upper bound helps control a as being close to $\|\mathbf{w}\|$, and lower keeps $t \lesssim d^C$
2. $t \lesssim \log \log(d) \eta^{-1}$
 - length of phase I
3. $\log(d) \ll p$ helps to ensure $\mathcal{E}_{\text{init}}$
 - so norm of $\mathbf{w}_\perp$ controlled, and balance of S^+, S^-
4. $\theta \ll \log^{-5C}(d)$
 - limited by the analysis of the per-step change in $\mathbf{w}_\perp$
5. $m \gg d \log^{7C}(d)$
 - Limited by Hoeffding's

Inductive Hypothesis

1. $\mathbf{w}_{\text{sp}}^{(t)} - \mathbf{w}_{\text{sp}}^{(t-1)} = \eta(1 \pm o(\log^{-2}(d))) \cdot (\frac{1}{2} - \lambda) \cdot \text{sgn}(a^{(0)}) (|\mathbf{w}_{\text{sp}}^{(t-1)}| + \theta)$

2. $\left\| \mathbf{w}_{\text{sig}}^{(t)} - \mathbf{w}_{\text{sig}}^{(t-1)} \right\| \lesssim \eta \left(\left\| \mathbf{w}_{\text{sig}} \right\|^{(t-1)} + \theta \log^C(d) d^{-1/2} \right)$
3. $\left\| \mathbf{w}_{\text{opp}}^{(t)} - \mathbf{w}_{\text{opp}}^{(t-1)} \right\| \lesssim \eta \left(\left\| \mathbf{w}_{\text{opp}} \right\|^{(t-1)} + \theta \log^C(d) d^{-1/2} \right)$
4. $\left\| \mathbf{w}_{\perp}^{(t)} - \mathbf{w}_{\perp}^{(t-1)} \right\| \lesssim \eta \theta \log^{-2C}(d)$
5. $\left\| \mathbf{w}_{\perp}^{(t)} - \mathbf{w}_{\perp}^{(t-1)} \right\|_{\infty} \lesssim \eta \theta \log^{3C}(d) d^{-1/2}$
6. $|a^{(t)}| \leq \left\| \mathbf{w}^{(t)} \right\|$

Phase I Result

1. $\left\| \mathbf{w}_{\text{sp}}^{(t)} \right\| \lesssim \theta \log^C(d)$
2. $\left\| \mathbf{w}_{\text{sig}} \right\|, \left\| \mathbf{w}_{\text{opp}} \right\| \lesssim \theta \log^{2C}(d) d^{-1/2}$
3. $\left\| \mathbf{w}_{\perp}^{(t)} \right\| \asymp \theta, \left\| \mathbf{w}_{\perp}^{(t)} - \mathbf{w}_{\perp}^{(0)} \right\| \lesssim \theta \log^{-C}(d)$
4. $\left\| \mathbf{w}_{\perp}^{(t)} \right\|_{\infty} \lesssim \theta \log^{4C}(d) d^{-1/2}$
5. $\left\| \mathbf{w}^{(t)} \right\| \asymp \left\| \mathbf{w}_{\text{sp}}^{(t)} \right\| + \left\| \mathbf{w}_{\perp}^{(t)} \right\| \lesssim \theta \log^C(d)$